

3 Computer Networks

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Learning Objectives:

- Understand IP and Routing.
- Understand how reliable data transmission is achieved via TCP/IP.
- Know what HTTP and DNS are.

3.1 Information Transmission

Information transmission is the process of sending information from one endpoint to another. In computer systems, the information to be transmitted is typically represented as a sequence of bits. Unlike human-readable representations such as Morse code, bits provide a simple and universal representation of information, while the interpretation of these bits can be left to upper layers.

There are two fundamental approaches to transmitting information: wired communication and wireless communication. In wired communication, physical media such as copper cables and optical fibers are used to carry signals between endpoints. Bits are transmitted as electrical signals over copper cables or as pulses of light over optical fibers. The data is encoded or modulated onto the physical signal at the sender and recovered at the receiver. In wireless communication, information is transmitted using electromagnetic waves through the air. Common examples include Wi-Fi, Bluetooth, cellular networks (4G/5G), and satellite communication. Digital data is modulated onto a carrier wave at the sender, and the receiver demodulates the signal to recover the original data. Different modulation methods can encode information in different properties of the carrier signal, such as its amplitude or frequency.

For example, in amplitude modulation (AM), the amplitude of the carrier signal is varied according to the information signal, while its frequency remains approximately constant. In frequency modulation (FM), the frequency of the carrier signal is varied according to the information signal, while its amplitude remains approximately constant.

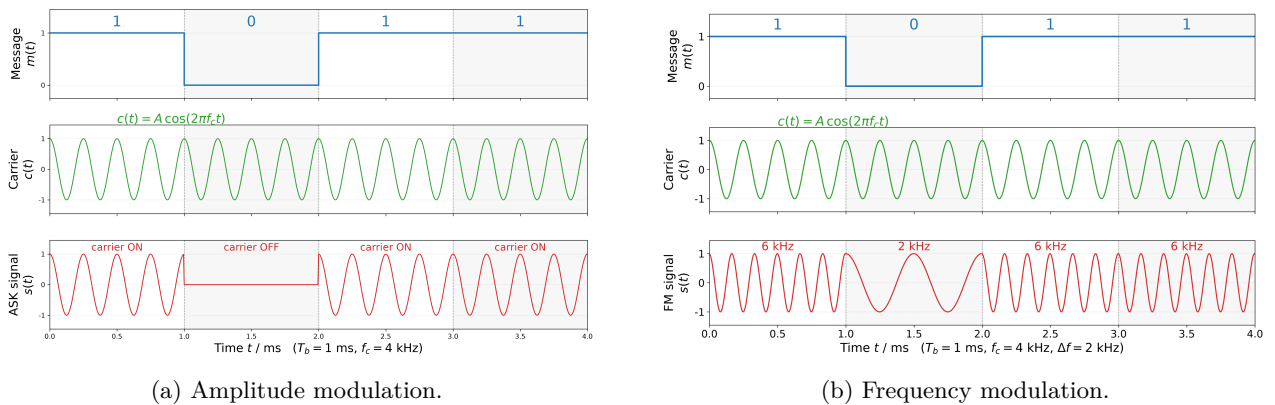


Figure 3.1: Modulation of the binary message “1011”.

3.2 Internet

The Internet is a network of networks. It is the global infrastructure that connects billions of devices, including computers, smartphones, servers, and sensors. Networks can be classified by their geographic scope. A Local Area Network (LAN) connects devices within a limited area, such as a home, office, or campus, while a Wide Area Network (WAN) connects networks over a larger geographic area. The Internet connects LANs and WANs into a global network.

3.2.1 MAC Address

Every device that connects to a network contains a Network Interface Controller (NIC). Each NIC is assigned a Media Access Control (MAC) address, which is a globally unique identifier used within local networks. You can think of a MAC address like a “citizen ID” for a piece of hardware. A standard MAC address is 48 bits long. The first 24 bits are assigned by the IEEE to a specific manufacturer, so all NICs produced by that manufacturer share the same prefix. The remaining 24 bits are assigned by the manufacturer to individual devices, making each NIC’s full address unique. This structure allows network devices to automatically distinguish one another on a local area network without any central registry.

MAC addresses are designed for local network communications. When data leaves a local network and travels across the Internet, MAC addresses are stripped off and replaced at each hop.

3.2.2 IP Address and Networks

The Internet Protocol (IP) is the fundamental addressing system that allows devices to locate and communicate with each other over Internet. Unlike MAC addresses, which are used within a local network, IP addresses are designed for end-to-end communication across multiple networks. Think of IP as the “street address” for computers: it tells data packets where to go so they can reach the correct destination.

An IP address (version 4) is 32 bits long, and it is usually written as four decimal numbers separated by dots, *e.g.*, 192.168.1.1. It consists of two parts:

- *Network prefix*: The left n bits that identify the network.
- *Host part*: The remaining bits that identify a specific device (host) on that network.

To determine the network part of an IP address, we use either the CIDR notation or subnet masks. The *CIDR notation* is a slash followed by the number of prefix bits. For example, 192.168.1.1/24 means the first 24 bits identify the network. Similarly, *subnet masks* specifies which bits are network bits. For instance, 192.168.1.1/255.255.255.0 also means the first 24 bits are the network part.

All IP addresses within the same network must share the same network prefix. For example, 192.168.1.1/24 and 192.168.1.100/24 are the same network because their first 24 bits are both 192.168.1.X, and there are 2^8 IP addresses within the network. 192.168.1.1/26 and 192.168.1.100/26 are not in the same network, because their first 26 bits differ, and the network capacity is only 2^6 .

$$\begin{array}{rcl} 192.168.1.1/24 & = & \underbrace{11000000.10101000.00000001}_{\text{Network Prefix}} | \underbrace{00000001}_{\text{Host part}} \\ 192.168.1.100/24 & = & \underbrace{11000000.10101000.00000001}_{\text{Network Prefix}} | \underbrace{01100100}_{\text{Host part}} \end{array}$$

Every IP network has a gateway, which is the device that connects the local network to other networks or the Internet. Figure 3.2 demonstrates a typical network with three devices, *i.e.*, a smartphone, a laptop,

and a Wi-Fi access point (AP). These devices are configured with IP addresses 192.168.1.1-192.168.1.3, and the gateway is the IP of the Wi-Fi AP, 192.168.1.1.

3.2.3 Address Resolution Protocol

While IP addresses are used for logical addressing, actual communication on a local network happens via MAC addresses. The Address Resolution Protocol (ARP) is a network protocol used to map a device's IP address to its MAC address within a local network.

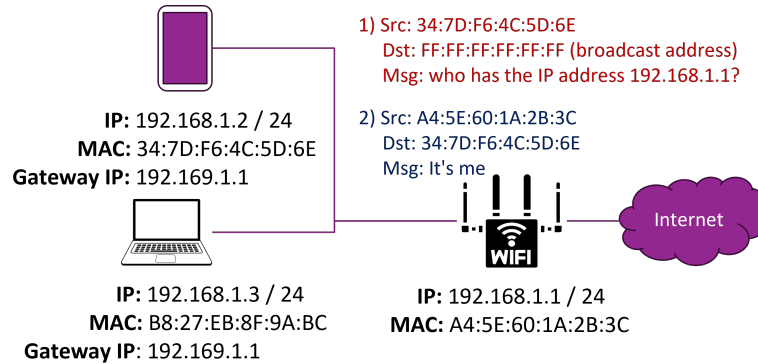


Figure 3.2: Demonstration of a local area network and ARP.

Figure 3.2 illustrates a scenario in which a smartphone (IP address: 192.168.1.2) queries the MAC address of the gateway (IP address: 192.168.1.1). We assume the smartphone already knows the gateway's IP address, as it is typically configured at the same time as the device's own IP address. However, it does not yet know the gateway's MAC address. To obtain it, the smartphone sends a broadcast message to all devices on the local network using the special MAC address FF:FF:FF:FF:FF:FF. This broadcast ensures that all devices on the local network receive the inquiry. The Wi-Fi access point recognizes that the requested IP address matches its own (192.168.1.1) and replies to the smartphone with its MAC address: A4:5E:60:1A:2B:3C.

3.2.4 Routing

When sending a message to a remote device, the packet needs to be routed. Routing is the process of determining how data packets are forwarded from the source to the destination across one or more networks. Unlike local communication within the same network, where a device can deliver a frame directly to the destination using its MAC address, communication across multiple networks requires intermediate devices, called *routers*, to forward packets toward their destination.

A router typically has multiple network interfaces, which allow it to communicate with different networks. Each interface is configured with one or more IP addresses that are valid on the network to which the interface is connected. For example, the Wi-Fi AP in Figure 3.3 acts as a router between the local network and the wide area network. Its internal interface is configured with the IP address 192.168.1.1/24, while its external interface is configured with the IP address 58.40.90.117/24. When a packet arrives, the router examines the destination IP address and consults its routing table to determine the appropriate outgoing interface and, when necessary, the next-hop router. The routing table specifies which destinations are reachable through which interface or next-hop router.

Now, let's put everything together. Suppose a message is sent from the host 192.168.1.2 to the destination host 61.152.229.228. The process is as follows:

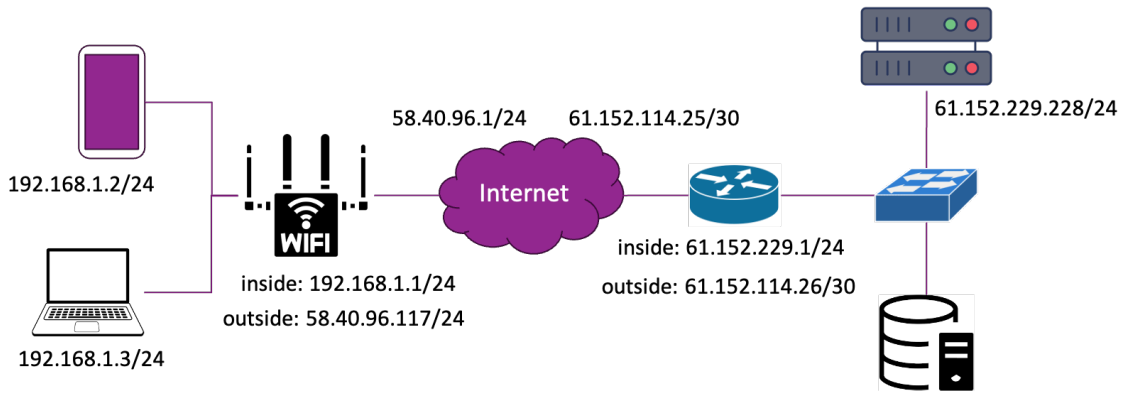


Figure 3.3: Demonstration of Wide Area Network and routing.

- 1) The sender compares the destination IP address with its own network prefix and determines that the destination is on a different network.
- 2) The sender uses ARP to obtain the MAC address of its gateway.
- 3) The sender sends the message to the gateway using the gateway's MAC address on the local network.
- 4) The gateway examines the destination IP address and consults its routing table to determine the next hop, 58.40.96.1, and forwards the message through its external interface.
- 5) The message travels through multiple routers across the Internet until it reaches a router connected to the destination network 61.152.229.0/30.
- 6) The destination router uses ARP to obtain the MAC address of the destination host and sends the message to 61.152.229.228 through its internal interface.

3.3 Messaging

3.3.1 OSI Model

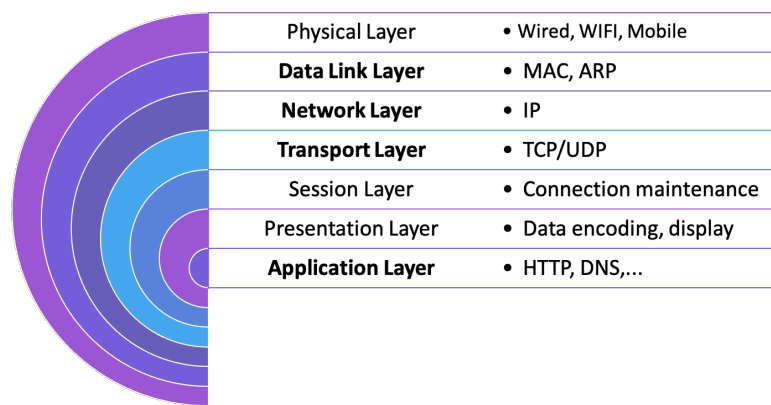


Figure 3.4: Demonstration of the OSI model.

In real-world scenarios, the OSI (Open Systems Interconnection) model defines seven conceptual layers to standardize network communication, as shown in Figure 3.4. Each layer has a specific role in processing, transmitting, and delivering data, from the software application down to the physical medium.

This layered approach allows different hardware and software systems to communicate seamlessly, while clearly separating responsibilities such as application services, transport reliability, logical addressing, local delivery, and physical transmission.

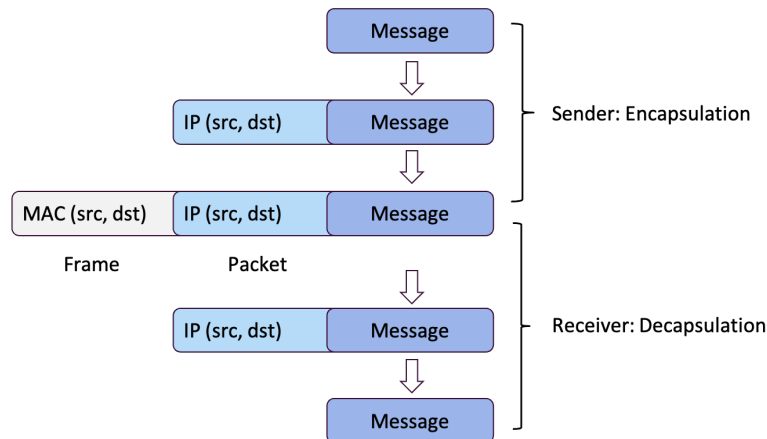


Figure 3.5: Demonstration of data encapsulation and decapsulation.

We have already introduced the MAC and IP which belong to the data link layer and network layer, respectively. To better understand how a message is sent using both MAC and IP addresses, Figure 3.5 provides a clear layered demonstration. The original data is first encapsulated with an IP header, which contains the source and destination IP addresses. Next, it is encapsulated with the source and destination MAC addresses. The MAC addresses are used for communication within the same local network, allowing devices to deliver frames directly to each other. Once the packet reaches a router or gateway, the MAC addresses are replaced for the next hop, while the IP addresses remain unchanged to guide the packet to its final destination.

3.3.2 Transmission Control Protocol

In practice, a network packet has a length limit, usually determined by the Maximum Transmission Unit (MTU) of the network, which is typically around 1500 bytes for Ethernet. If a message is larger than this limit, it cannot be sent in a single packet and must be segmented into multiple packets. A big problem arises concerning the integrity of the message: when a message is split into multiple packets, there is a risk that some packets may be lost, duplicated, or arrive out of order during transmission. Without a mechanism to handle these issues, the receiver might reconstruct the message incorrectly or miss parts of it entirely. This is where Transmission Control Protocol (TCP) plays a crucial role. TCP ensures reliable, ordered delivery by:

- *Sequencing*: Assigning a sequence number to each segment so the receiver can reorder them correctly.
- *Acknowledgment*: The receiver sends an acknowledgment for every segment received, letting the sender know it arrived safely.
- *Retransmission*: If an acknowledgment is not received within a certain time, the sender retransmits the segment.
- *Error Checking*: TCP uses checksums in each segment to detect corruption, ensuring that only valid data is accepted.

By combining these mechanisms, TCP guarantees that the original message is reassembled accurately at the destination, preserving both its order and integrity, even over unreliable networks. Next, we employ Figure 3.6 to demonstrate how the first three designs are achieved.

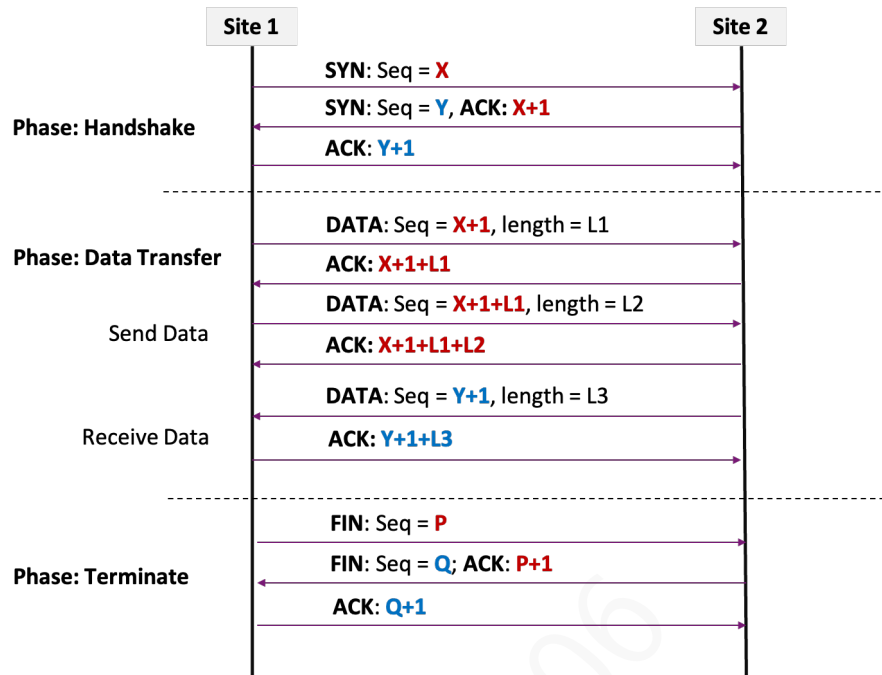


Figure 3.6: Demonstration of TCP.

There are three major phases in a TCP connection:

1) **Handshake:** This phase establishes a reliable connection between the sender and receiver. TCP uses a *three-way handshake*:

- The client sends a SYN packet with a random sequence number X to the server to request a connection.
- The server responds with a SYN-ACK packet with its own random sequence number Y and acknowledges the client's sequence number X by setting the acknowledgment number to $X+1$, indicating it expects the next byte to be $X+1$.
- The client sends an ACK packet with acknowledgment number $Y+1$ back to the server, completing the handshake.

After this phase, both sides know each other's sequence numbers and are ready to exchange data.

2) **Data Transfer:** In this phase, the actual message is transmitted. TCP divides the message into smaller segments, assigns a sequence number to each segment, and sends them reliably. Each sequence number is calculated as the previous sequence number plus the length of the segment in bytes. The receiver acknowledges each segment by sending an ACK with the sequence number of the next expected segment. If a segment is lost, corrupted, or received out of order, the sender retransmits it, ensuring that all segments are delivered in order and without errors.

3) **Connection Termination:** Once the communication is complete, TCP gracefully closes the connection to free resources. This typically involves a *four-way handshake*, although it may sometimes be completed in three messages when an ACK and FIN are combined.:

- One side sends a **FIN** packet to indicate it has finished sending data.
- The other side acknowledges with an **ACK**.
- The second side then sends its own **FIN** packet.
- Finally, the first side acknowledges with an **ACK**, completing the termination.

3.3.3 Network Applications

Above the transport layer, there are different applications. The most popular ones are Domain Name System (DNS) and Hypertext Transfer Protocol (HTTP). Application-layer protocols define the actual structure and meaning of the messages exchanged between programs. HTTP is the foundation of the World Wide Web. It enables browsers and servers to exchange web pages, images, and data through a request–response model. A client, such as a browser, sends a request, and the server replies with the requested content or an error status code. DNS acts like the Internet’s “phone book”. Humans remember host names like `www.example.com`, but routers need IP addresses. DNS translates these names into IP addresses by querying a hierarchy of distributed name servers. Together, DNS and HTTP show how application protocols build on the transport layer to deliver real services to users.

Exercise

- 1) Bob is connecting to his friend's network. The gateway IP address is `211.100.2.128/25`, while Bob's IP address is configured as `211.100.2.100/25`. Are they on the same network? Can Bob access the Internet through this gateway?
- 2) Hands-on Exercise: Wireshark is a free, open-source protocol analyzer that captures and displays packets moving across a network. Download and install Wireshark from its official website at <https://www.wireshark.org>. Launch the program, select your active network interface, and start a live capture. Then access a chosen website in your browser and observe the traffic captured by Wireshark.
 - Check the sequence numbers and acknowledgements of TCP segments to understand reliable delivery.
 - Identify the HTTP or HTTPS packets generated when the page loads.
 - Filter for DNS packets and note the query and response for the website's domain name.